



Utkarsh Rai

Berlin, Germany +49-15207591796 utkarsh26rai@gmail.com

LinkedIn: www.linkedin.com/in/utkarshrai-ds **Github:** <https://github.com/UtkarshRai-ds>

Profile

Aspiring Junior Data Scientist with practical experience in machine learning, computer vision, deep learning, and data analysis. Demonstrated ability to deliver value by designing and deploying data pipelines, developing predictive models, and creating interactive dashboards. Skilled at generating data-driven insights that produce measurable results.

Education

2023 – 2025
Berlin, Germany

Data Analytics | MSc
Berlin School of Business and Innovation

2019 – 2023
Greater Noida, India

Business Analytics & Marketing | BBA+MBA Integrated
Sharda University

Work Experience

09/2024 – 10/2025
Berlin, Germany

Data Science Freelancer
Self Employed

- Loaded, preprocessed, and augmented image datasets using TensorFlow, then designed and trained CNN-based classifiers achieving high accuracy on client-specific use cases.
- Extracted and parsed local data from PDFs and spreadsheets with Python, automating contact-list ingestion scripts.
- Fine-tuned existing ML models (scikit-learn, Keras, TensorFlow) through hyperparameter optimization and feature engineering, increasing predictive performance by up to 20%.
- Engineered end-to-end ETL pipelines for ingesting, cleaning, and transforming structured and unstructured data.
- Measured and monitored pipeline performance to ensure reliability and efficiency.
- Created SQL-based ETL workflows, reducing data processing and load times by 30% while boosting reporting reliability.

06/2024 – 10/2024
Noida, India

Data Analytics Intern (Remote)
APEX Byte

- Implemented data-quality checks (missing values, duplicates, outliers) using pandas and stored procedures, achieving > 99% data integrity for downstream analytics.
- Wrote and optimized complex SQL queries and indexed key tables, cutting average query runtime by 27% and accelerating insights delivery.
- Partnered with product and marketing teams to translate business requirements into data models and KPIs, enabling data-driven decision-making across the organization.

07/2022 – 09/2022
Delhi, India

Business Analytics Intern
Goodspace Technologies Pvt. Ltd.

- Developed and validated linear and logistic regression models to predict user churn, improving retention strategies by 15%.
- Performed data mining and exploratory analysis on 100k+ user events using Python (pandas, NumPy), that increased conversion rate by 12%.
- Modelled datasets using Power BI and DAX, improving data accuracy by 30%.

Projects & Publications

05/2026

Telco Customer Churn Prediction

Projects & Publications

- Built an end-to-end ML pipeline on the IBM Telco dataset (7,043 rows × 21 features), engineering 36 features through domain knowledge, statistical analysis, and interaction transformations using pandas and scikit-learn.
- Tuned Logistic Regression with Optuna (30 trials, 5-fold CV), achieving AUC 0.838 and Recall 0.80 targeting the top 51% risk tier captures 87% of churners, effectively translating model output into a practical business retention strategy.
- Deployed a 4-page Streamlit dashboard using Docker, GitHub Actions CI/CD, MLflow experiment tracking.
- telco-churn-utkarsh.streamlit.app.

01/2025

Emoji Prediction System

- Developed an emoji prediction system leveraging natural language processing techniques to enhance user communication experience.
- Processed and analyzed large-scale textual datasets to train the system, optimizing feature extraction for contextual understanding.

10/2024

Skywatch-Satellite Object Detection

- Designed and trained dual YOLOv8 models (YOLOv8n and YOLOv8s) on the NWPU VHR-10 satellite imagery dataset (800 images, 10 classes, 4,811 annotations) utilizing Weights & Biases for experiment tracking.
- Achieved mAP50 of 0.697 and Recall of 0.690 with YOLOv8s; identified a bridge detection failure (0.03 mAP50 despite 29.5% class share) and analyzed root causes including extreme aspect ratios and label ambiguity.
- Delivered a 5-page interactive Streamlit dashboard featuring live inference, dual-model comparison, and per-class AP analysis; deployed via Docker and GitHub Actions CI.
- huggingface.co/spaces/Utkarsh-DS/Skywatch-Detection

11/2023

Predictive Analytics in E-Commerce Sales Forecasting and Customer Segmentation

- Used SQL and data transformation tools to create segmentation models, design A/B tests, and inform campaign planning for improving customer conversion and retention.

11/2022

Presented a paper on the topic “MetaVerse: the new shape of marketing” in the International Marketing Conference hosted by Indian Institute of Management Shillong

02/2021

Book chapter published in “Emerging Contours of Business and Management” on topic “PHARMEASY: A fast growing healthtech venture” ISBN: 978-93-91842-41-3

Certifications

04/2024

Generative AI for Data Scientist

04/2023

Introduction to Statistics, Stanford University, Coursera

04/2023

PowerBI for Business Intelligence, Microsoft, Coursera

01/2023

Data Science Professional Certificate from IBM Skills, Coursera

Languages

English: C1 level

German: A2 level – actively improving to B1

Hindi: Native

Additional Information

Eligible to work in Germany with a post-study work permit (18-month Job Seeker Visa valid from September 2025). Open to relocate.